

Cinderblock Contract Billing Requirements

Developer handoff and business logic source of truth

Standard Invoice, Progress Invoice, Change Orders, Retainage, Payment Schedule, and QuickBooks Online synchronization

Status Final functional requirements

Updated September 20, 2026

Audience Frontend, backend, QA, product, design, and future maintainers

Functional behavior only. This document does not prescribe application architecture.

Executive Summary

An Accepted Estimate establishes one Contract for tracked billing. Progress Invoices consume accepted source scope from that Contract, and Accepted Change Orders revise the same Contract without resetting billing history. Retainage withholds part of an invoice amount due without reducing completed work. Payment Schedule plans milestone billing but never overrides Contract availability. QuickBooks Online receives ordinary accounting documents while Cinderblock remains authoritative for Contract billing state.

Standard Invoice is a separate full-scope copy/conversion path. It remains linked to its source Estimate and copies applicable terms, but it does not consume tracked Contract Progress or source-line availability. Standard Invoice and Progress Invoice share Billed as the completed Contract status; only the Progress path can pass through Partially Billed.

Accepted Estimate	Current Contract	Progress billing	Payments and Retainage release
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Interpretation Rule

Interpretation rule: If behavior is not explicitly defined in these requirements, do not infer new business behavior from incidental prototype UI. The prototype illustrates approved workflows but does not independently define new business rules.

Explicit requirements override illustrative screenshots.

Authority and Precedence

1. Newer explicit product decisions override older requirements.
2. The canonical Contract billing business rules and financial specification override conflicting feature-prototype wording.
3. The QBO synchronization requirements control the accounting integration boundary and post-sync safeguards.
4. Approved implemented behavior may clarify an existing rule, but a prototype shortcut does not create a new rule.
5. A genuine unresolved business contradiction must appear in Open Questions and Conflicts Found rather than being silently decided during implementation.

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1 Canonical Terminology	Exact product terms used throughout
2 Global Contract Model	Contract state, financial measures, lifecycle, and pools
3 Standard Invoice	Flexible copy conversion outside tracked Progress consumption
4 Progress Invoice	Tracked Contract billing and customer presentation
5 Change Order	Accepted Contract revisions and approval lifecycle
6 Retainage	Withholding, held balance, and release invoices
7 Payment Schedule	Advisory milestones and invoice relationships

Section	Purpose
8 QuickBooks Online Sync	Accounting document mapping and conflict safeguards
9 State Tables	Availability, editing, and Contract effects
10 Business Invariants	Rules that must always remain true
11 Cross Feature Interaction Matrix	Effects of major actions
12 Validation and Warnings	Hard blocks, warnings, and informational states
13 Customer Facing Documents	Cross-document output rules
14 Internal UI Business Rules	Visible business state and locks
15 Canonical Numerical Examples	Reconciled Before Action After examples
16 Edge Cases	Important non-happy flows
17 Open Questions and Conflicts Found	Unresolved production business decisions
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19 Technical Details	Short implementation constraints required for correctness
20 Final Self Audit	Consistency and integrity result

1 Canonical Terminology

Term	Definition
Estimate	A proposal document that may be accepted and may become the source of a Contract.
Accepted Estimate	The accepted revision that establishes the Contract for tracked billing.
Contract	The accepted commercial scope used for tracked billing. It is the Accepted Estimate plus all Accepted Change Orders.
Original Contract	The Accepted Estimate before the effect of Accepted Change Orders.
Current Contract	The current accepted Contract after applying all Accepted Change Orders.
Gross Contract Scope	Accepted base line scope plus Cost Plus, before discount, tax, retainage, and payments.
Contract subtotal	Gross accepted Contract scope before discount. Where Cost Plus is explicit accepted scope, it is included.
Contract Value	Gross Contract Scope less accepted contractual discount, excluding tax, retainage, and payments.
Contract Progress	Posted Gross Contract Scope invoiced divided by current Gross Contract Scope.
Standard Invoice	A flexible invoice copied from an Estimate that does not consume tracked Contract Progress.
Progress Invoice	A tracked invoice that reserves and posts accepted Contract source scope.
Contract Invoice	The source-grouped presentation variant of the same Progress Invoice model after Accepted Change Orders exist.

Term	Definition
Change Order	An Estimate-style document that proposes a signed revision to the existing Contract.
Retainage	Discounted principal withheld from an invoice's current Amount Due without reducing completed scope.
Retainage Balance	The Contract-level amount retained and not yet released, subject to active Draft reservations.
Retainage release	An Invoice that makes previously withheld Retainage due without billing new work.
Payment Schedule	An advisory list of billing milestones and their invoice relationships.
Milestone	One planned or actual billing event within Payment Schedule.
Remaining subtotal	Gross, pre-discount Contract source scope available for a new Progress Invoice after posted use and Draft reservations.
Previously Billed	Posted tracked billing preceding the current invoice in the immutable billing sequence.
This Invoice	The current invoice's gross allocation to a Contract line or computed Cost Plus fee.
Balance to Finish	Contract Amount less posted and current allocations for the line, subject to current accepted scope.
Percent complete	Cumulative attributed billing divided by the positive Contract Amount of a line.
Discount	A document amount that reduces the represented principal according to applicable rules.
Discount Pool	The finite Contract-level discount established by the Accepted Estimate and signed Accepted Change Order adjustments.
Tax	Sales tax calculated outside Contract scope and value using the applicable tax configuration.
Tax Credit Balance	Historical tax attributable to removed accepted scope and available only to offset tax on future normal invoices.
Cost	Internal contractor cost. It never appears in customer output and does not determine Cost Plus.
Cost Plus	An accepted percentage term calculated from customer-facing selling Price.
Cost Plus Fee	The computed amount of Cost Plus scope, tracked with its base scope and presented to customers as one aggregate invoice item.
Draft Invoice	A saved invoice that reserves applicable capacity but is not posted.
Open Invoice	A posted invoice whose allocations and financial effects are recognized.
Accepted	For Contract billing, the Contract is accepted with no posted tracked billing. For a Change Order, it separately records approval.
Partially Billed	Tracked billing exists and the Current Contract still has uninvoiced scope.
Billed	The Current Contract has no remaining gross scope, or an active Standard Invoice has completed the full-scope copy path.
Linked Estimate	The Estimate relationship displayed on an Invoice or Change Order for document provenance.
Linked Invoice	An invoice relationship displayed from an Estimate, Change Order, or Payment Schedule.

2 Global Contract Model

The following atomic rules define the shared Contract, invoice posting, financial measures, pools, and lifecycle behavior used by every feature.

GLOBAL-001 Accepted Estimate establishes the Contract

Accepting an Estimate establishes the Contract used for tracked Progress billing.

GLOBAL-002 Current Contract composition

Before any Accepted Change Order, the Contract equals the Accepted Estimate. After acceptance, each Change Order revises that same Contract, so the Current Contract equals the Accepted Estimate plus all Accepted Change Orders.

GLOBAL-003 One Progress accounting model

Progress Invoicing uses one Contract-based accounting model from Estimate acceptance onward. The first Accepted Change Order does not create a second model, migrate history, or reset any ledger.

GLOBAL-004 Source line lineage

Every accepted Estimate and Change Order source line retains its source document, source line identity, accepted quantity and amount, posted allocation, Draft reservation, and remaining billable allocation.

GLOBAL-005 No Estimate to Change Order line mapping

The product must not require a one-to-one mapping between an Estimate line and a Change Order line. Removed and replacement scope are independent accepted adjustments with preserved lineage.

GLOBAL-006 Gross Contract Scope

Gross Contract Scope equals accepted base line scope plus applicable Cost Plus scope, before discount, tax, retainage, and payments.

GLOBAL-007 Contract Value

Contract Value equals Gross Contract Scope less the accepted Contract Discount. Tax, retainage, and payments do not change Contract Value.

GLOBAL-008 Nonnegative Contract Value

Estimate or Change Order acceptance must fail without mutation when the resulting Contract Value would be below zero. Contract Value may equal zero.

GLOBAL-009 Contract Progress

Contract Progress equals posted Gross Contract Scope invoiced divided by current Gross Contract Scope. Discount, tax, retainage, and payments do not increase or decrease progress.

GLOBAL-010 Total Invoiced

Total Invoiced equals posted base and Cost Plus allocations less discount applied to posted invoices. Tax, retainage, and payments are excluded.

GLOBAL-011 Remaining subtotal

Remaining subtotal is the gross, pre-discount Contract source scope available for a new Progress Invoice after posted billing and active Draft reservations. The Discount Pool is not subtracted.

GLOBAL-012 Contract summary Remaining

The Contract summary Remaining equals Contract Value less Total Invoiced. It is a different net measure from Remaining subtotal.

GLOBAL-013 Accepted Progress state

The tracked Contract state is Accepted when the Contract is accepted and no tracked invoice has posted.

GLOBAL-014 Partially Billed state

The Contract state is Partially Billed when posted tracked billing exists and current gross Contract scope remains uninvoiced.

GLOBAL-015 Billed state

The Contract state is Billed when posted tracked billing exists and no current gross Contract scope remains available under the completion rules.

GLOBAL-016 Billed Contract reopening

Accepting new positive scope on a Billed Contract changes its billing state to Partially Billed when prior posted billing remains and the revised Contract has uninvoiced scope.

GLOBAL-017 Change Order lifecycle independence

Draft, Pending, Accepted, Declined, and Canceled describe Change Order approval history. They remain distinct from the Contract's billing state.

GLOBAL-018 Invoice posting event

Open is the posting event. A Draft reserves selected scope and financial pools but does not contribute to Previously Billed, Total Invoiced, Contract Progress, or recognized retainage.

GLOBAL-019 Invoice lifecycle

The tracked invoice lifecycle is Draft to Open to Partially Paid to Paid. An Open invoice cannot return to Draft.

GLOBAL-020 One active tracked Draft

At most one active tracked invoice Draft may reserve a Contract at a time, including a Progress Invoice Draft or Retainage release Draft.

GLOBAL-021 Latest invoice financial eligibility

Only the latest active tracked Open invoice may be financially edited or canceled, and only when ordinary payment prerequisites allow it. A later Draft or accepted Contract revision locks earlier Open invoices financially.

GLOBAL-022 Payment activity remains available

A financial lock does not block ordinary payment application, void, refund, or unapply behavior. Payment actions do not reverse Contract allocations or restore edit eligibility to an earlier invoice.

GLOBAL-023 Draft release behavior

Deleting or canceling a Draft releases its reserved base scope, Cost Plus, discount, retainage release capacity, and Tax Credit, and releases its active milestone association.

GLOBAL-024 Posted cancellation behavior

Canceling an eligible posted tracked invoice reverses its posted allocations and financial pool consumption, preserves history and document number, and recalculates Contract and Payment Schedule state.

GLOBAL-025 Discount Pool

The Contract Discount Pool combines the Accepted Estimate discount and signed discount adjustments from Accepted Change Orders. Draft, Pending, Declined, and Canceled Change Orders do not change it.

GLOBAL-026 Discount reservations

A Draft reserves its selected nonnegative discount. Opening consumes it, and Draft deletion or eligible invoice cancellation releases or restores it.

GLOBAL-027 Unused discount

A contractor may invoice all gross scope without exhausting the Discount Pool. Opening the final gross-scope invoice with unused discount requires a warning but remains allowed.

GLOBAL-028 Cost Plus basis

Cost Plus is calculated from customer-facing selling Price, not internal Cost. The fee is explicit accepted scope with lineage to the priced source line.

GLOBAL-029 Cost Plus consumption

An invoice consumes Cost Plus scope in the same proportion as its associated base scope and must not calculate Cost Plus again over the invoice subtotal.

GLOBAL-030 Tax boundary

Tax is outside Gross Contract Scope, Contract Value, Contract Progress, and Total Invoiced. Contract-derived invoices inherit the accepted Contract tax configuration and component taxability.

GLOBAL-031 Tax Credit Balance

Historical tax attributable to accepted deductive scope creates a Tax Credit Balance that may offset only calculated tax on future normal invoices. It never reduces principal or creates an automatic refund.

GLOBAL-032 Retainage boundary

Retainage is withheld from discounted principal, excluding tax. It changes Amount Due and Accounts Receivable but not Gross Contract Scope, Total Invoiced, Contract Progress, or milestone consumption.

GLOBAL-033 Payment Schedule boundary

Payment Schedule is an advisory planning and history layer based on Contract Value. It never reserves Contract scope or overrides invoice availability rules.

GLOBAL-034 Historical allocations

Accepted Contract revisions never redistribute historical billing. Posted amounts remain attributed to the source lines and document revisions on which they occurred.

GLOBAL-035 Source of truth

Cinderblock remains authoritative for Contract billing state. External accounting edits must not recalculate Cinderblock scope, progress, pools, ledgers, milestones, or lineage.

2 Global Contract Model Requirements Index

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ID	Requirement
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GLOBAL-031	Tax Credit Balance
GLOBAL-032	Retainage boundary

ID	Requirement
GLOBAL-033	Payment Schedule boundary
GLOBAL-034	Historical allocations
GLOBAL-035	Source of truth

Canonical Contract Flow

Estimate accepted	Contract established	COs revise Contract	Progress invoices consume scope	Payments and releases settle A R
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3 Standard Invoice

Standard Invoice is the flexible copy path. It is Contract-derived for provenance and copied terms, but it remains outside tracked Contract Progress consumption.

STDINV-001 Purpose

A Standard Invoice copies Estimate scope into a flexible invoice when the contractor does not need tracked Progress billing.

STDINV-002 Contract relationship

A Standard Invoice remains linked to its source Estimate and uses the copied Estimate terms, but it does not participate in the tracked Contract Progress allocation model.

STDINV-003 Eligibility

Standard Invoice creation is available only before a tracked Progress Invoice or active Standard Invoice conversion has made the billing paths mutually exclusive.

STDINV-004 Entry choice

For an eligible Estimate with no invoice history, Create invoice offers Standard invoice and Progress invoice. Selecting Standard invoice opens the unsaved Standard Invoice editor without an additional setup step.

STDINV-005 Copied lines

Every Estimate line is copied into the Standard Invoice with Item name, Price, Qty, and Amount.

STDINV-006 Flexible editing

Copied Standard Invoice lines may be edited or removed, and ordinary invoice lines may be added under existing Standard Invoice rules.

STDINV-007 No progress columns

The Standard Invoice does not display Contract Amount, Previously Billed, Balance to Finish, or percent complete.

STDINV-008 No Contract consumption

Saving, editing, opening, or paying a Standard Invoice does not consume tracked Contract source-line availability or update tracked Contract Progress.

STDINV-009 Billed status timing

The source Estimate becomes Billed only after the Standard Invoice Draft saves successfully. Opening and abandoning an unsaved editor does not change the Estimate's status.

STDINV-010 Exclusive conversion lock

While an active Standard Invoice exists, tracked Progress Invoice creation, Change Order creation, and the applicable Payment Schedule path are unavailable for the source Estimate.

STDINV-011 Conversion lock recovery

Deleting the Standard Invoice Draft or canceling the Standard Invoice removes the conversion lock when ordinary history, payment, and conflict rules allow it.

STDINV-012 Payment Schedule confirmation

When a Payment Schedule exists, choosing Standard invoice requires confirmation that the full Estimate will be converted and the schedule removed.

STDINV-013 Payment Schedule transaction boundary

Production removes the Payment Schedule only when Standard Invoice creation succeeds. Canceling setup or abandoning the unsaved editor preserves the schedule.

STDINV-014 Saved navigation

Saving creates a numbered Draft and keeps the user on the saved Standard Invoice. Returning to the Estimate requires explicit close or navigation.

STDINV-015 Linked Estimate

The Standard Invoice displays Linked to Estimate for provenance. The relationship does not create progress tracking or synchronization between copied lines and Estimate lines.

STDINV-016 Cost Plus aggregate

When copied scope includes Cost Plus, the invoice contains one computed aggregate Cost plus fee entry. It must not create one fee row per underlying line.

STDINV-017 Cost Plus fee identity

The computed fee uses item name Cost plus fee and description Cost plus at {percent}. It is not independently editable.

STDINV-018 Customer Cost Plus presentation

On the customer PDF and Web Link, Cost plus fee appears once after regular items and before Subtotal, and its amount is included in Subtotal.

STDINV-019 No duplicate customer fee

When Cost Plus is presented as a customer line item, the totals section must not also show Cost plus percentage or Cost plus fee rows.

STDINV-020 Conditional customer fee

When Cost Plus does not apply, no Cost plus fee line or zero-value Cost Plus placeholder appears.

STDINV-021 Customer content

Customer PDF and Web Link use the invoice's customer, job, dates, lines, public note, attachments, terms, status, payment information, and applicable discount and tax.

STDINV-022 Preview availability

PDF preview and Customer web preview are hidden in every unsaved New Invoice state and appear in their existing location only after the invoice is saved.

3 Standard Invoice Requirements Index

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STDINV-014	Saved navigation
STDINV-015	Linked Estimate
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STDINV-017	Cost Plus fee identity
STDINV-018	Customer Cost Plus presentation
STDINV-019	No duplicate customer fee
STDINV-020	Conditional customer fee
STDINV-021	Customer content
STDINV-022	Preview availability

Standard Invoice Happy Flow

Eligible Estimate	Copy to Invoice	Edit Standard Invoice	Save Draft	Estimate Billed
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4 Progress Invoice

Progress Invoice is the tracked billing path. It consumes accepted Contract scope while preserving source-line history, reservations, and cumulative progress.

PROGINV-001 Purpose

A Progress Invoice bills part or all of an Accepted Contract while preserving Contract-level totals and source-line billing history.

PROGINV-002 Contract basis

Every Progress Invoice bills the Current Contract. With no Accepted Change Order the Contract consists of the Accepted Estimate; Accepted Change Orders revise the same Contract.

PROGINV-003 Entry condition

Progress Invoice creation requires an Accepted Contract with eligible remaining source scope and no conflicting active tracked Draft.

PROGINV-004 First invoice choices

The first Progress Invoice offers Percent or amount and Select items. It does not offer a rest-of-estimate option.

PROGINV-005 Subsequent rest choice

A subsequent Progress Invoice offers The rest of this estimate before Accepted Change Orders and The rest of this contract after Accepted Change Orders.

PROGINV-006 Percent or amount

Percent or amount supports mutually exclusive percentage and dollar entry modes. Percentage is based on the original Contract basis and is limited by the remaining percentage.

PROGINV-007 Remaining subtotal label

The Progress Invoice creation workflow labels the gross available amount Remaining subtotal. It does not relabel unrelated Remaining values elsewhere.

PROGINV-008 Remaining subtotal percentage

In subsequent percentage mode, the workflow displays Remaining subtotal with both the dollar limit and maximum remaining percentage. Dollar mode displays the dollar limit only.

PROGINV-009 Input validation

Blank, nonnumeric, zero, negative, and over-available inputs cannot create an invoice. An over-available entry shows Exceeds remaining subtotal and retains the entered value for correction.

PROGINV-010 Selected item behavior

Select items shows accepted source lines, keeps fully invoiced lines visible but disabled, and bills the full remaining amount of each selected eligible line.

PROGINV-011 Selection count

Select all and selection counts include only eligible lines. The primary action remains unavailable when no eligible line is selected.

PROGINV-012 All lines in editor

Every accepted Contract source line appears in the resulting editor, including unselected, superseded, removed, and fully invoiced lines where required for historical context.

PROGINV-013 Internal columns

The internal tracked invoice table displays Item name, Qty, Contract Amount, Previously Billed, This Invoice, Balance to Finish, and percent complete where the approved surface calls for those measures.

PROGINV-014 Previously Billed definition

Previously Billed includes posted tracked allocations from invoices that precede the current invoice in the immutable billing sequence. It excludes the current invoice, later invoices, Drafts, and payments.

PROGINV-015 This Invoice

This Invoice is the editable current allocation for an eligible Draft or eligible latest unpaid Open invoice. It cannot exceed source-line or Contract availability.

PROGINV-016 Percent complete

For an ordinary positive line, percent complete equals Previously Billed plus This Invoice divided by Contract Amount. It is derived and not independently editable.

PROGINV-017 Removed line presentation

A negative Change Order adjustment is non-billable and displays an em dash for Previously Billed, This Invoice, and percent complete. Historical billing remains on the originating line.

PROGINV-018 Apply remaining

An editable billable line with unused capacity offers Apply remaining. The control disappears at the maximum and never appears on locked, removed, non-billable, fully billed, or customer-facing rows.

PROGINV-019 Draft reservation

Saving the Draft atomically reserves its base scope, associated Cost Plus, selected discount, retainage release capacity where applicable, and Tax Credit.

PROGINV-020 Posting

Changing Draft to Open converts reservations into posted allocations, records recognized retainage, consumes reserved discount and Tax Credit, and updates Contract Progress and history.

PROGINV-021 Single Draft guard

When an active tracked Draft exists, the product must not create another Progress Invoice or Retainage release Draft against the Contract.

PROGINV-022 Discount step

When Discount Pool remains, the final creation step asks Apply discount, suggests a proportional amount, permits zero, and blocks an amount above the available pool.

PROGINV-023 Discount scope separation

Progress Invoice scope selection uses gross source availability. Discount is selected separately and never reduces the displayed Remaining subtotal.

PROGINV-024 Tax inheritance

A Progress Invoice inherits the accepted Contract tax configuration and component taxability. Its Draft stores the resolved tax facts, and opening does not rerate them.

PROGINV-025 Tax editor lock

The tracked invoice editor displays inherited tax information as read-only and does not provide an independent rate or taxability override.

PROGINV-026 Tax Credit application

A Draft reserves no more than the lesser of calculated invoice tax and available Tax Credit Balance. The credit appears as a separate negative totals row and never reduces principal.

PROGINV-027 Cost Plus aggregate

Cost Plus appears internally as one aggregate computed fee associated with all applicable billed base scope, not as one fee row per underlying line.

PROGINV-028 Cost Plus identity

The aggregate fee uses item name Cost plus fee and description Cost plus at {percent}. The inherited percentage and computed amount are not independently editable.

PROGINV-029 Cost Plus tracking

Internal tracking preserves Contract Cost Plus fee, Previously Billed Cost Plus fee, Cost Plus billed on this invoice, and Remaining Cost Plus fee.

PROGINV-030 Cost Plus row fields

The aggregate Cost Plus row populates Contract Amount, Previously Billed, and This Invoice from the associated base-scope allocation.

PROGINV-031 Cost Plus percent complete

The Cost Plus fee row leaves percent complete blank. The product must not calculate or display a synthetic completion percentage for the fee.

PROGINV-032 Customer Cost Plus line

On the customer PDF and Web Link, one Cost plus fee line appears after regular invoice items and before Subtotal, with description Cost plus at {percent}. Subtotal includes the fee.

PROGINV-033 No duplicate customer Cost Plus

Customer totals do not repeat Cost plus percentage or Cost plus fee rows when the fee is presented in the items table.

PROGINV-034 Customer progress columns

Customer output includes Item name and This Invoice and may include Qty, Contract Amount, Previously Billed, Balance to Finish, percent complete, and Taxed according to the approved per-invoice Customer view settings.

PROGINV-035 Unbilled rows

Items not billed controls only customer presentation. When included, zero-current rows are visually subdued; hiding them does not change invoice allocations.

PROGINV-036 Fractional quantity presentation

Customer output may hide or simplify fractional quantities according to existing presentation rules without changing the actual stored billed quantity.

PROGINV-037 Invoice History

Starting with the second tracked invoice, customer PDF and Web Link show prior tracked invoices in chronological order and exclude the current invoice.

PROGINV-038 Invoice History amount

Invoice History Amount is each prior invoice's actual total after its applicable discount and tax, not its gross Contract allocation.

PROGINV-039 Contract Progress customer block

Customer output shows cumulative Contract Progress after current invoice totals when the contract-aware presentation applies. It does not expose internal pools or reservations.

PROGINV-040 Saved navigation

Creating the invoice keeps the user on the numbered Draft. Closing an unsaved editor discards its work without changing the Contract; closing a saved invoice returns by explicit navigation.

PROGINV-041 Preview availability

PDF preview and Customer web preview are hidden before the invoice exists and become available only after the Draft has been saved.

PROGINV-042 Change Order acceptance collision

Accepting a Change Order while a tracked Draft exists cancels that Draft, releases its reservations, preserves it read-only in history, and requires a new invoice against the revised Contract.

PROGINV-043 Accepted revision lock

Accepting a Contract revision financially locks earlier Open tracked invoices. New or changed scope must be billed on a new invoice and cannot be added retroactively.

PROGINV-044 Final unused discount warning

Opening the invoice that consumes the final gross scope while discount remains available shows the unused discount and resulting over-invoiced amount, permits continuation, and applies no discount automatically.

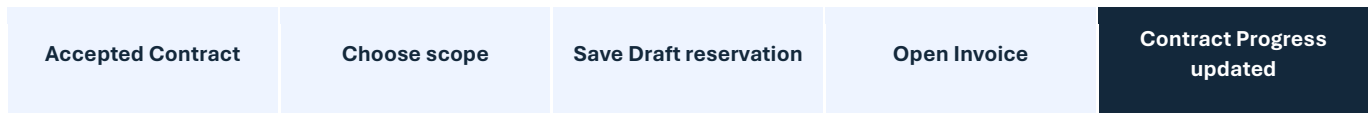
4 Progress Invoice Requirements Index

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PROGINV-043	Accepted revision lock
PROGINV-044	Final unused discount warning

Progress Invoice Happy Flow



5 Change Order

A Change Order proposes a signed Contract revision while preserving its own approval history and the Contract's existing billing history.

CO-001 Purpose

A Change Order is a separate Estimate-style document that proposes independent additions, removals, discount adjustments, or zero-value contract changes.

CO-002 Contract relationship

An Accepted Change Order revises the existing Contract linked to the original Estimate. It does not establish a second Contract.

CO-003 Numbering

Each saved Change Order receives the next nonreused contract-level identifier in the form {EstimateID}-CO1, {EstimateID}-CO2, and so on.

CO-004 Creation origins

Change Order creation may begin from the Accepted Estimate or an eligible Accepted Change Order while preserving the original Estimate relationship.

CO-005 Standard Invoice lock

An active Standard Invoice conversion disables Change Order creation for its source Estimate.

CO-006 Parallel proposals

Multiple Draft or Pending Change Orders may coexist when they reserve unrelated source lines. A source line may participate in only one active Draft or Pending Change Order at a time.

CO-007 Conflict presentation

A line reserved by another Draft or Pending Change Order remains visible but disabled with This item already has an open Change Order and a reference to the open document.

CO-008 Valid compositions

Addition-only, removal-only, mixed, named zero-value, and real net-zero Change Orders are valid when they contain a meaningful adjustment.

CO-009 Removal representation

Selected existing scope is represented as a normal negative-quantity line with positive Price, negative Qty, and signed Amount. It remains independent of any replacement line.

CO-010 Partial quantity correction

A partial quantity change removes the original accepted quantity and adds an ordinary new line at the corrected quantity. The product does not create replacement pairing.

CO-011 Added scope

New scope uses ordinary editable Item name, Price, Qty, and Amount fields and has no data relationship to a removed line.

CO-012 Internal Cost excluded

Change Order editing, summaries, and customer output do not show or calculate from internal Cost or Markup.

CO-013 Tax authority

A Change Order does not establish an invoice tax rate. Applicable future invoices inherit tax configuration from the accepted Contract.

CO-014 Cost Plus lock

A Change Order inherits the accepted Contract Cost Plus percentage as read-only and cannot introduce, remove, or modify that rate.

CO-015 Cost Plus adjustments

Additions and full-negative removals create signed Cost Plus scope using the inherited percentage. Other economic adjustments use ordinary Change Order lines.

CO-016 Discount adjustment

A Change Order may explicitly increase or decrease contractual discount. A positive discount adjustment increases the Discount Pool; a negative adjustment reduces it but may not take the pool below zero.

CO-017 Discount exception

A Change Order may add a discount greater than the previously available Discount Pool because it changes the contractual pool. This exception does not apply to Progress Invoice discount consumption.

CO-018 Draft state

Saving a meaningful New Change Order creates a Draft, assigns its number, and keeps the Draft open for editing.

CO-019 Pending state

Sending a saved Draft or selecting Pending creates the Pending state. Pending has no accepted Contract effect and continues to reserve participating source lines.

CO-020 Acceptance revision

Acceptance uses the current saved revision. Dirty financial changes must be saved before acceptance, and a stale customer link cannot accept a newer revision.

CO-021 Accepted immutability

An Accepted Change Order is financially immutable, never returns to Pending, and cannot be Declined, Canceled, or deleted. Corrections require another Change Order.

CO-022 Declined and Canceled

Declined and Canceled Change Orders have no Contract effect, remain historical, and release their open source-line reservations.

CO-023 Acceptance Contract effect

Acceptance adds the Change Order's signed base scope, Cost Plus scope, and discount adjustment to the Current Contract without rewriting earlier billing.

CO-024 Future invoice availability

Future Progress Invoice availability reflects the revised accepted scope while prior posted allocations remain attributed to their original source lines.

CO-025 Negative Contract boundary

Acceptance is blocked when the proposed Change Order would make Contract Value negative. The product must explain the condition without clamping or partially accepting the document.

CO-026 Over-invoicing after deduction

When an Accepted deductive Change Order reduces accepted scope below posted billing, the product shows the over-invoiced condition and does not automatically create a refund, credit, or negative invoice.

CO-027 Tax Credit generation

Accepted removal of previously taxed scope generates Tax Credit from persisted historical tax attribution. It does not change the Change Order delta or Contract Progress.

CO-028 Contract Details

Saved Contract surfaces show Original contract, each Change Order in sequence, and Current contract. Current contract includes Accepted deltas only.

CO-029 Contract summary

Saved Change Orders show Contract Value, Total Invoiced, Remaining or Over invoiced, and Contract Progress. Draft and Pending deltas appear as proposed context without changing accepted totals.

CO-030 Linked invoices

Linked to is reserved for invoices associated with the specific Estimate or Change Order and displays only the invoice link identity, not status or amount.

CO-031 Customer PDF and link

Every saved Change Order provides a PDF and Customer Web Link by reusing the Estimate customer document model.

CO-032 Customer identity

The customer document title is Change Order, shows the Change Order number, and shows Linked to Estimate {Estimate number}.

CO-033 Customer content

The customer document uses the Change Order's own items, subtotal, applicable discount, applicable Cost Plus, total, public note, attachments, terms, customer and job information, and acceptance status.

CO-034 Conditional customer totals

Discount, tax, and Cost Plus fields are absent from Change Order customer totals when they do not apply. Zero-value placeholders are not shown.

CO-035 Delete eligibility

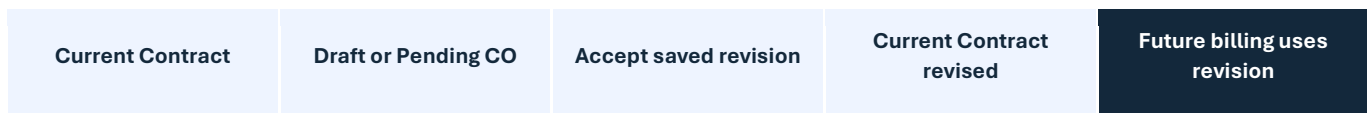
Draft, Pending, Declined, and Canceled Change Orders may be deleted after confirmation. Accepted Change Order deletion is blocked.

5 Change Order Requirements Index

ID	Requirement
CO-001	Purpose
CO-002	Contract relationship
CO-003	Numbering
CO-004	Creation origins
CO-005	Standard Invoice lock
CO-006	Parallel proposals
CO-007	Conflict presentation
CO-008	Valid compositions
CO-009	Removal representation
CO-010	Partial quantity correction
CO-011	Added scope
CO-012	Internal Cost excluded
CO-013	Tax authority
CO-014	Cost Plus lock
CO-015	Cost Plus adjustments
CO-016	Discount adjustment
CO-017	Discount exception

ID	Requirement
CO-018	Draft state
CO-019	Pending state
CO-020	Acceptance revision
CO-021	Accepted immutability
CO-022	Declined and Canceled
CO-023	Acceptance Contract effect
CO-024	Future invoice availability
CO-025	Negative Contract boundary
CO-026	Over-invoicing after deduction
CO-027	Tax Credit generation
CO-028	Contract Details
CO-029	Contract summary
CO-030	Linked invoices
CO-031	Customer PDF and link
CO-032	Customer identity
CO-033	Customer content
CO-034	Conditional customer totals
CO-035	Delete eligibility

Change Order Happy Flow



6 Retainage

Retainage is a standalone optional billing layer. Withholding changes Amount Due; release invoices make held principal due without billing new scope.

RET-001 Purpose

Retainage withholds part of a tracked invoice's currently due discounted principal while preserving full gross billing against Contract scope.

RET-002 Optional feature

Retainage appears only for Contracts where the feature is configured. Ordinary Contracts do not show Retainage fields, totals, actions, or zero placeholders.

RET-003 Estimate default

The Estimate may define an optional Retainage percentage for future tracked invoices. Blank means no Retainage.

RET-004 Historical stability

Changing the Estimate default affects only future invoice Drafts. Saved invoice rates and historical withheld and released amounts never recalculate.

RET-005 Invoice override

A Progress Invoice Draft copies the current default and permits a zero-through-100-percent override that applies only to that invoice.

RET-006 Calculation basis

Retainage is calculated against discounted principal, excluding tax. It does not withhold tax.

RET-007 Calculation order

Invoice totals apply billed base and Cost Plus scope, then discount, then tax on discounted taxable principal, then Retainage on discounted principal, and finally Total Due.

RET-008 No line allocation

Cinderblock treats Retainage as a totals-level amount. It does not create a line-level Retainage allocation or Retainage completion column.

RET-009 Progress unaffected

Retainage withheld does not reduce This Invoice, line completion, Contract Progress, Total Invoiced, or Payment Schedule fulfillment.

RET-010 Tracking visibility

The Retainage tracking block remains hidden while Total retained is zero and no release history exists. The configured percentage may remain visible before withholding.

RET-011 Total retained

Total retained is cumulative Retainage actually recognized on posted Progress Invoices and never decreases when Retainage is released.

RET-012 Released

Released increases only when a Retainage release Draft becomes Open.

RET-013 Held

Held equals Total retained minus Released.

RET-014 Available to release

Available to release equals Held minus the amount reserved by an active Retainage release Draft.

RET-015 Contract level pool

Retainage Balance is maintained once for the Contract. The product does not create separate line, milestone, invoice, or Change Order release pools.

RET-016 Release eligibility

Release retainage is available only when Available to release is greater than zero and no conflicting tracked Draft exists.

RET-017 Partial and full release

A release may be partial or full. The entered amount must be greater than zero and no greater than Available to release.

RET-018 Release Draft

Creating a release opens a New retainage invoice. Saving creates a numbered Draft and reserves the selected release amount without counting it as Released.

RET-019 Release content

A Retainage release Invoice contains one specialized Retainage release amount and does not resend contract work, source lines, milestones, or original quantities.

RET-020 Release effect

Opening a Retainage release Invoice makes previously withheld principal due. It does not create Contract scope, revenue, tax, Total Invoiced, Contract Progress, or milestone consumption.

RET-021 Release cancellation

Canceling or deleting the release Draft restores its reserved release capacity. Canceling a posted release follows ordinary invoice payment and cancellation rules.

RET-022 Customer withheld presentation

A customer Progress Invoice shows Retainage withheld clearly as a deduction from Amount Due.

RET-023 Customer release presentation

A release PDF and Web Link identify Retainage release as previously withheld money being made due and must not imply new work.

RET-024 Customer release totals

Release customer totals begin with Total and then show actual Amount Paid where applicable and Balance Due. They do not repeat a Retainage released totals row.

RET-025 Invoice History

Retainage release history remains visible without changing the prior Progress Invoice amounts or Contract billing history.

6 Retainage Requirements Index

ID	Requirement
RET-001	Purpose
RET-002	Optional feature

ID	Requirement
RET-003	Estimate default
RET-004	Historical stability
RET-005	Invoice override
RET-006	Calculation basis
RET-007	Calculation order
RET-008	No line allocation
RET-009	Progress unaffected
RET-010	Tracking visibility
RET-011	Total retained
RET-012	Released
RET-013	Held
RET-014	Available to release
RET-015	Contract level pool
RET-016	Release eligibility
RET-017	Partial and full release
RET-018	Release Draft
RET-019	Release content
RET-020	Release effect
RET-021	Release cancellation
RET-022	Customer withheld presentation
RET-023	Customer release presentation
RET-024	Customer release totals
RET-025	Invoice History

Retainage Happy Flow

Progress Invoice	Retainage withheld	Held balance	Release Invoice	Previously withheld amount due
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7 Payment Schedule

Payment Schedule is an advisory planning and history layer. It guides invoice creation but never creates or overrides Contract availability.

PAY-001 Purpose

Payment Schedule plans and records billing milestones. A Progress Invoice executes billing and remains the authority for Contract consumption.

PAY-002 Location

Payment Schedule belongs to the Estimate and Contract workflow and does not create a separate Contract or financial ledger.

PAY-003 Calculation basis

Planned milestone amounts are based on Contract Value, including accepted Cost Plus and discount and excluding tax, Retainage, and payments.

PAY-004 Milestone fields

Each managed milestone requires a name and may include a percentage or equivalent amount plus an optional planned date.

PAY-005 Editable percentage and amount

Percentage and Amount appear together and remain editable. Percentage entry accepts at most three decimal places; Amount entry stores currency precision and derives a precise percentage anchor.

PAY-006 Amount authority

The stored currency-precision milestone Amount is authoritative for billing defaults, schedule totals, and validation. Calculations must not reconstruct it from a rounded displayed percentage.

PAY-007 Zero Contract Value

At zero Contract Value, percentage milestones remain valid and project zero. Dollar entry is unavailable until a meaningful percentage can be derived.

PAY-008 Preset behavior

Company Settings presets contain percentage-based milestones so they can apply at zero and recalculate as Estimate scope changes.

PAY-009 Planning does not reserve

Creating, editing, reordering, or overallocating milestones does not reserve Contract scope, discount, Tax Credit, or Retainage.

PAY-010 Over-allocation warning

When milestone amounts exceed current Contract Value, the product shows Payment schedule exceeds 100% in Manage milestones and on the schedule table.

PAY-011 Over-allocation remains editable

Over-allocation is informational. It does not disable Save, force correction, normalize rows, rebalance future milestones, or expand invoice availability.

PAY-012 Milestone selection

When unfulfilled milestones exist, Progress Invoice creation offers Select milestone before ad-hoc scope methods.

PAY-013 Milestone invoice method

The milestone screen offers Percent or amount and Select items. A planned percentage is an editable suggestion rather than a billing restriction.

PAY-014 Milestone Contract limits

An invoice created from a milestone remains subject to Contract availability, source-line availability, Discount Pool, Draft reservation, and overbilling rules.

PAY-015 Gross default

Milestone creation defaults gross allocation to the milestone percentage of Gross Contract Scope.

PAY-016 Discount suggestion

Milestone creation suggests the same percentage of the applicable original Contract Discount, capped at the remaining Discount Pool.

PAY-017 Actual amount

The milestone's actual is the invoice's resulting net contract billing after the contractor's selected discount. It may differ from the planned amount.

PAY-018 One invoice association

A milestone may have one active invoice association. Multiple draws for one phase require multiple milestones.

PAY-019 Planned state

A Planned milestone has no active invoice and may offer Create invoice.

PAY-020 Draft invoice state

A Draft invoice milestone shows its linked invoice and provisional amount with neutral treatment and does not offer another Create invoice action.

PAY-021 Invoiced state

A milestone becomes Invoiced only when its associated Draft becomes Open.

PAY-022 Canceled association

Deleting or canceling a Draft, or validly canceling the latest eligible Open invoice, returns the live milestone to Planned while preserving immutable invoice history.

PAY-023 Dates

Planned date and invoice date are stored separately. Display Planned: {date} only when present and Invoiced: {date} when an invoice date exists.

PAY-024 No sequential lock

The schedule does not restrict invoice creation to the next milestone. Visual emphasis on the expected next milestone does not disable later eligible milestones.

PAY-025 Linked milestone editing

Once linked, Percentage and Amount are historical and disabled. The name, planned date, and order retain their established edit behavior.

PAY-026 Rename behavior

Renaming a Draft-linked milestone updates the Draft without warning. Renaming an issued milestone requires confirmation; Paid invoices retain their historical milestone name.

PAY-027 Reordering

Reordering moves the complete stable milestone record and does not change invoice associations or historical invoice snapshots.

PAY-028 Change Order recalculation

After an Accepted Change Order changes Contract Value, unfulfilled milestone amounts recalculate from their precise percentage anchors. Historical invoiced amounts do not change.

PAY-029 Actual variance

An actual invoice that differs from plan updates that milestone actual and does not redistribute or rebalance other milestones.

PAY-030 Ad hoc invoices

Every saved ad-hoc Progress Invoice creates or updates one milestone. A blank optional name becomes Invoice X using the row's one-based insertion position.

PAY-031 Standard Invoice interaction

Choosing Standard Invoice with a schedule requires confirmation and removes the schedule only when invoice creation succeeds.

PAY-032 Retainage interaction

Retainage reduces Amount Due but does not reduce milestone consumption. A Retainage release does not create, consume, reopen, or modify a milestone.

PAY-033 On acceptance setting

Before acceptance, a schedule may choose Do nothing or Create draft invoice for milestone 1. The setting is hidden after acceptance.

PAY-034 Automatic Draft guard

Automatic Draft creation at acceptance uses milestone 1's full valid amount, is idempotent, and obeys the one-Draft-per-Contract guard without blocking Estimate acceptance.

PAY-035 Existing Draft conflict

When automatic creation encounters an existing Draft, acceptance succeeds, the existing Draft remains unchanged, the milestone remains unconsumed, and the product shows an inline error.

7 Payment Schedule Requirements Index

ID	Requirement
PAY-001	Purpose

ID	Requirement
PAY-002	Location
PAY-003	Calculation basis
PAY-004	Milestone fields
PAY-005	Editable percentage and amount
PAY-006	Amount authority
PAY-007	Zero Contract Value
PAY-008	Preset behavior
PAY-009	Planning does not reserve
PAY-010	Over-allocation warning
PAY-011	Over-allocation remains editable
PAY-012	Milestone selection
PAY-013	Milestone invoice method
PAY-014	Milestone Contract limits
PAY-015	Gross default
PAY-016	Discount suggestion
PAY-017	Actual amount
PAY-018	One invoice association
PAY-019	Planned state
PAY-020	Draft invoice state
PAY-021	Invoiced state
PAY-022	Canceled association
PAY-023	Dates
PAY-024	No sequential lock
PAY-025	Linked milestone editing
PAY-026	Rename behavior
PAY-027	Reordering
PAY-028	Change Order recalculation
PAY-029	Actual variance
PAY-030	Ad hoc invoices
PAY-031	Standard Invoice interaction
PAY-032	Retainage interaction

ID	Requirement
PAY-033	On acceptance setting
PAY-034	Automatic Draft guard
PAY-035	Existing Draft conflict

Payment Schedule Happy Flow

Contract and schedule	Select milestone	Create Progress Draft	Open Invoice	Milestone records actual
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8 QuickBooks Online Sync

These rules describe only the Contract Billing behavior introduced or materially affected at the QBO boundary. Existing general QBO customer, item, trigger, and lifecycle behavior remains authoritative.

QBO-001 System ownership

Cinderblock owns Contract billing logic and remains the source of truth. QBO is an accounting destination.

QBO-002 No QBO Progress Invoicing

The integration must not use QBO Progress Invoicing or maintain QBO Estimate-to-Invoice progress relationships.

QBO-003 Internal concepts excluded

QBO does not receive Contract Progress, Previously Billed, Balance to Finish, Remaining subtotal, pools, ledgers, milestone state, Change Order relationships, reservations, or Contract revisions.

QBO-004 Estimate mapping

A Cinderblock Estimate synchronizes to a standard QBO Estimate.

QBO-005 Change Order mapping

Each Cinderblock Change Order synchronizes to a separate standard QBO Estimate and does not modify the original QBO Estimate.

QBO-006 Change Order identity

The integration preserves the Cinderblock Change Order identity in the QBO transaction mapping without requiring a QBO relationship to the original Estimate.

QBO-007 Standard Invoice mapping

A Cinderblock Standard Invoice synchronizes to a standard QBO Invoice without an Estimate-to-Invoice link.

QBO-008 Progress Invoice mapping

A Cinderblock Progress or Contract Invoice synchronizes to a standard QBO Invoice containing ordinary accounting lines and no Cinderblock Contract lineage.

QBO-009 Retainage release mapping

A Retainage release synchronizes to a standard QBO Invoice that may contain only one positive Retainage release line.

QBO-010 Actual quantities

QBO receives the actual billed quantity, rate, amount, taxability, and mapped Product or Service or account for each invoice line.

QBO-011 Fractional quantities

A fractional Progress Invoice quantity must synchronize even when Cinderblock hides or simplifies that quantity in customer presentation.

QBO-012 Retainage withheld line

A nonzero withheld amount appends one QBO line named Retainage with Qty 1 and the persisted amount as negative.

QBO-013 Retainage release line

A release synchronizes one QBO line named Retainage release with Qty 1 and the persisted amount as positive.

QBO-014 No repeated work on release

A Retainage release must not resend contract work, Estimate lines, Change Order lines, milestones, or original quantities.

QBO-015 Discount representation

When a discount appears on the Cinderblock document, synchronize it through the established compatible QBO representation. When absent, do not send a zero or synthetic discount.

QBO-016 Cost Plus representation

When Cost Plus appears on the Cinderblock document, synchronize the represented amount without asking QBO to reproduce the calculation. When absent, do not send a zero or synthetic fee.

QBO-017 Tax alignment

For a QBO-connected account, synchronize taxability and line facts so Cinderblock totals align with the connected QBO tax configuration and rate.

QBO-018 Local operation independence

A valid Cinderblock operation remains valid when QBO synchronization fails. QBO failure must not roll back or silently rewrite the Cinderblock document.

QBO-019 Last successful sync baseline

The integration retains the QBO company and transaction identity, type, concurrency value, last success time, synchronized Cinderblock revision, material accounting snapshot or fingerprint, and last outcome.

QBO-020 Material field coverage

The divergence baseline covers document identity, customer, dates, lines, quantities, rates, amounts, taxability, tax, discounts, Retainage, Cost Plus, totals, status, void or deletion state, and payment-sensitive state when available.

QBO-021 Preflight

Before updating, voiding, deleting, or recreating a mapped transaction, Cinderblock compares current QBO state with the last successfully synchronized state.

QBO-022 Optimistic concurrency

Every resolved QBO write uses the current QBO concurrency value. A race detected after preflight creates a conflict rather than an automatic overwriting retry.

QBO-023 Unchanged transaction

When QBO still matches the last successful state, a later Cinderblock edit may use the normal update path with current version protection.

QBO-024 Edited transaction

When QBO was materially edited after sync, Cinderblock preserves both states, marks a sync conflict, explains the mismatch, and requires intentional resolution before another material write.

QBO-025 Deleted or voided transaction

When QBO reports the mapped transaction deleted, voided, or unavailable, Cinderblock does not silently recreate it and requires intentional reconciliation.

QBO-026 Payment sensitive transaction

When a QBO Invoice has an applied payment, credit, deposit, or settlement state, Cinderblock does not blindly replace, delete, void, or materially rewrite it.

QBO-027 Cinderblock cancellation

Deleting, voiding, canceling, or invalidating a Cinderblock document does not automatically delete its mapped QBO transaction. The product inspects divergence, status, and payments first.

QBO-028 Create or update failure

A failed QBO create or update leaves the Cinderblock transaction intact, records an actionable failure, and never presents successful synchronization.

QBO-029 Retry safety

A retry is idempotent when safe. An ambiguous timeout requires lookup or request-identity reconciliation before retry so it cannot create a duplicate.

QBO-030 Conflict state

Sync conflict is a first-class integration state separate from document lifecycle and does not reopen, unpost, or recalculate the Cinderblock document or Contract.

QBO-031 Review changes

Review changes compares current QBO state with both the last-synchronized and current Cinderblock representations.

QBO-032 Keep QBO guardrail

Keep QuickBooks version must not import QBO edits into Cinderblock Contract ledgers. Its accounting-document and future-baseline behavior requires an explicit approved rule.

QBO-033 Overwrite guardrail

Overwrite QuickBooks requires permission, confirmation, audit, and a fresh concurrency and payment preflight, and is unavailable when accounting conditions make it unsafe.

QBO-034 Audit trail

Conflict detection, review, choice, actor, timestamps, versions, and outcome are auditable.

QBO-035 Sync statuses

The product distinguishes Not synced, Sync pending or in progress, Synced successfully, Sync failed, and Sync conflict.

QBO-036 Safe unresolved default

Until an explicit resolution rule exists, preserve both systems, block the risky QBO write, surface the mismatch, and require intentional manual resolution.

QBO-037 Billing status mapping

Cinderblock Partially Billed and Billed both map to QBO's existing Converted semantic. The integration preserves the distinct Cinderblock statuses and must not infer remaining Contract scope from QBO Converted.

8 QuickBooks Online Sync Requirements Index

ID	Requirement
QBO-001	System ownership
QBO-002	No QBO Progress Invoicing
QBO-003	Internal concepts excluded
QBO-004	Estimate mapping
QBO-005	Change Order mapping
QBO-006	Change Order identity
QBO-007	Standard Invoice mapping
QBO-008	Progress Invoice mapping
QBO-009	Retainage release mapping
QBO-010	Actual quantities
QBO-011	Fractional quantities
QBO-012	Retainage withheld line
QBO-013	Retainage release line
QBO-014	No repeated work on release
QBO-015	Discount representation
QBO-016	Cost Plus representation
QBO-017	Tax alignment

ID	Requirement
QBO-018	Local operation independence
QBO-019	Last successful sync baseline
QBO-020	Material field coverage
QBO-021	Preflight
QBO-022	Optimistic concurrency
QBO-023	Unchanged transaction
QBO-024	Edited transaction
QBO-025	Deleted or voided transaction
QBO-026	Payment sensitive transaction
QBO-027	Cinderblock cancellation
QBO-028	Create or update failure
QBO-029	Retry safety
QBO-030	Conflict state
QBO-031	Review changes
QBO-032	Keep QBO guardrail
QBO-033	Overwrite guardrail
QBO-034	Audit trail
QBO-035	Sync statuses
QBO-036	Safe unresolved default
QBO-037	Billing status mapping

9 State Tables

Estimate and Contract States

State	Meaning	Available actions
Draft	Estimate not accepted	Edit; manage schedule; send or accept under existing rules
Pending or Sent	Current saved revision awaits acceptance	Edit creates a new revision; accept current revision
Accepted	Accepted Contract with no posted tracked billing	Create eligible Standard or Progress Invoice; create CO; manage schedule
Partially Billed	Posted tracked billing exists and scope remains	Create Progress Invoice; create CO; manage schedule; release Retainage when available

State	Meaning	Available actions
Billed	Current gross scope is complete through posted tracked billing, or an active Standard Invoice exists	Progress path: no new invoice until positive scope is accepted; Standard path: tracked Progress Invoice, CO, and schedule unavailable

Change Order States

State	Contract effect	Financial editing	Destructive behavior
New	None	Editable	Close discards
Draft	Proposed only	Editable unless a valid dependency locks it	Delete with confirmation
Pending	Proposed only	Current revision awaits acceptance	Cancel, decline, or delete with confirmation
Accepted	Signed delta revises Current Contract	Immutable	Delete, decline, and cancel blocked
Declined	None	Historical	Delete allowed
Canceled	None	Historical	Delete allowed

Tracked Invoice States

State	Contract allocation	Financial editing	Payment activity
New	None	Editable working state	Unavailable
Draft	Reservations only	Editable if it is the active Draft	Not posted
Open	Posted	Only latest active unpaid tracked Open invoice	Available
Partially Paid	Posted	Financially locked	Available
Paid	Posted	Financially locked	Available under existing rules
Canceled	Reversed when eligible	Read-only history	Handled by existing cancellation and payment rules

Payment Schedule States

State	Invoice relationship	Presentation	Next action
Planned	None	Planned values	Create invoice may be available
Draft invoice	One active Draft	Neutral linked invoice and Draft badge	Open or cancel the Draft
Invoiced	One posted invoice	Fulfilled treatment and actual amount	No second invoice from that milestone
Returned to Planned	Prior invoice canceled	Live association removed; history preserved	Create invoice may be available again

10 Business Invariants

- INV-001** An Accepted Estimate establishes one Contract for tracked billing.
- INV-002** Every Progress Invoice bills an Accepted Contract.
- INV-003** Accepted Change Orders revise the same Contract without resetting history.
- INV-004** Contract availability is supported by accepted source-line availability.
- INV-005** Estimate and Change Order lines do not require one-to-one mapping.
- INV-006** A Progress Invoice cannot allocate more than the remaining Contract or source-line scope.
- INV-007** A Draft reserves capacity but does not become Previously Billed or Contract Progress.
- INV-008** A Progress Invoice cannot consume more discount than the available Discount Pool.
- INV-009** A Change Order may increase the contractual Discount Pool.
- INV-010** Cost Plus percentage cannot be changed by a Progress Invoice or Change Order.
- INV-011** Cost Plus is consumed once with associated base scope and never recomputed over an invoice subtotal.
- INV-012** Tax remains outside Contract scope and progress.
- INV-013** Retainage withheld does not reduce completed Contract scope.
- INV-014** Retainage release does not create Contract scope or a second tax event.
- INV-015** Payment Schedule never authorizes billing beyond Contract availability.
- INV-016** A Standard Invoice does not participate in tracked Contract consumption.
- INV-017** A positive Accepted Change Order can return a Billed Contract to Partially Billed.
- INV-018** Remaining subtotal is gross and does not subtract the Discount Pool.
- INV-019** Customer-facing documents never expose internal Cost.
- INV-020** QBO never becomes the source of Cinderblock Contract billing state.

11 Cross Feature Interaction Matrix

Action	Scope and value	Pools and tax	Progress and history	Schedule Retainage QBO
Accept Estimate	Establishes accepted scope and Contract Value	Establishes discount, Cost Plus, and tax basis	Contract state becomes Accepted	Enables plan; no QBO change unless normal Estimate sync runs
Create Standard Invoice	Copies Estimate; does not consume tracked scope	Copies represented terms	Creates separate Standard conversion lock	Removes schedule only on successful create; ordinary QBO Invoice
Save Progress Draft	Reserves selected base and Cost Plus scope	Reserves selected discount and Tax Credit	Does not change posted progress	Associates Draft milestone; may reserve release capacity; sync waits for existing policy
Open Progress Invoice	Consumes selected scope	Consumes discount and Tax Credit; posts stored tax	Increases Total Invoiced and Contract Progress	Milestone becomes Invoiced; Retainage becomes retained; ordinary QBO Invoice

Action	Scope and value	Pools and tax	Progress and history	Schedule Retainage QBO
Accept positive CO	Increases current accepted scope and value	May change discount; adds inherited Cost Plus	Preserves history; may change Billed to Partially Billed	Recalculates future milestones; separate QBO Estimate
Accept negative CO	Decreases current accepted scope and value	May generate Tax Credit; removes associated Cost Plus	Preserves history; may expose over-invoicing	Recalculates future milestones; separate QBO Estimate
Create Retainage release	No Contract scope or value change	No discount, new tax, or Cost Plus	No Total Invoiced or progress change	No milestone consumption; decreases held balance when Open; ordinary QBO Invoice
Cancel eligible invoice	Releases or reverses allocations	Releases or restores pools	Recalculates history without reusing number	Milestone returns to Planned; Retainage effects reverse; QBO requires safe reconciliation
Record payment	No scope or Contract Value change	No pool change	No Contract Progress change	Changes A/R only; may constrain QBO conflict resolution

12 Validation and Warnings

ID	Condition	Product result	Type
VALID-001	Aggregate overbilling	Block a Progress Invoice allocation above the Contract's available Remaining subtotal.	Hard block
VALID-002	Source line overbilling	Block a line allocation above that source line's available quantity or amount.	Hard block
VALID-003	Discount overuse	Block Progress Invoice discount below zero or above the available Discount Pool.	Hard block
VALID-004	Existing Draft	Block creation of a second tracked Draft and direct the user to the existing Draft.	Hard block
VALID-005	Fully invoiced Contract	Disable new tracked invoice creation when no accepted gross scope remains available.	Hard block
VALID-006	Change Order line conflict	Disable a source line already reserved by another Draft or Pending Change Order and identify the conflicting document.	Hard block
VALID-007	Meaningless Change Order	Block saving, sending, or accepting a Change Order that contains no meaningful adjustment.	Hard block
VALID-008	Negative Contract Value	Block Estimate or Change Order acceptance when the resulting Contract Value would be less than zero.	Hard block
VALID-009	Schedule over-allocation	Show Payment schedule exceeds 100% as a warning and permit editing and saving.	Warning
VALID-010	Blank milestone name	Block saving a manually managed Payment Schedule while any milestone name is blank or whitespace only.	Hard block
VALID-011	Invalid milestone allocation	Block a nonpositive invoice input and block any milestone invoice amount above available Contract scope.	Hard block
VALID-012	Invoice exceeds milestone	When a proposed invoice exceeds its milestone plan but remains within Contract availability, warn that the proposed amount becomes the milestone actual and allow continuation.	Warning
VALID-013	Retainage unavailable	Hide or disable Retainage release when no Available to release balance exists.	Hard block
VALID-014	Retainage over-release	Block a release amount at or below zero or above Available to release.	Hard block
VALID-015	Final unused discount	Warn, but do not block, when the final gross scope is being posted while	Warning

ID	Condition	Product result	Type
		Contract discount remains unused.	
VALID-016	Stale Contract write	Reject a stale Contract-affecting financial write without partial mutation and preserve unsaved editor state for the established retry or refresh path.	Hard block
VALID-017	Payment edit race	When invoice editing races with payment application, the first successful operation wins and the second fails cleanly against current state.	Hard block
VALID-018	Stale acceptance link	Reject customer acceptance when the link does not represent the current acceptable Estimate or Change Order revision.	Hard block
VALID-019	QBO divergence	Block a risky QBO write when the mapped transaction materially differs from the last successful sync and create a Sync conflict state.	Hard block
VALID-020	QBO retry ambiguity	Do not retry an ambiguous QBO create or update until lookup or idempotency evidence prevents duplication.	Hard block

13 Customer Facing Documents

Document	Identity and relationship	Items and totals	Additional content
Estimate	Estimate title and number	Estimate items; applicable discount, Cost Plus, tax, subtotal, and total	Customer and job information, note, attachments, terms, acceptance
Standard Invoice	Invoice number and Linked to Estimate	Ordinary items plus one Cost plus fee item when applicable; fee included before Subtotal	Dates, status, note, attachments, terms, payments
Progress Invoice	Invoice number, Linked Estimate, optional milestone	Progress columns under visibility settings; one Cost plus fee item; discount, tax, Retainage, totals	Contract Progress and prior Invoice History when applicable
Change Order	Change Order title and number; Linked to Estimate	CO items and its own applicable discount, Cost Plus, subtotal, tax treatment, and total	Customer and job information, note, attachments, terms, acceptance
Retainage release	Invoice number, Retainage release context, Linked Estimate	One Retainage release item; Total, Amount Paid where applicable, Balance Due	Dates, note, terms, payment history

Tax, Discount, Cost Plus, and Retainage appear only when applicable. A customer document never shows zero-value placeholder rows or internal Cost, Markup, margin, profit, pools, reservations, or ledger mechanics.

14 Internal UI Business Rules

UI requirements appear here only when visibility, editing, locking, or warning treatment communicates a business state.

UI-001 Conditional totals

Across Estimate, Invoice, Progress Invoice, and Change Order totals, Tax, Discount, and Cost Plus rows are absent when the value does not apply. The product does not show zero-value placeholders.

UI-002 Internal Cost privacy

Internal Cost, Markup, margin, and profit never appear in customer-facing documents or links.

UI-003 Locked field clarity

A field locked by a business rule must be visibly distinguishable from an enabled editor control and must provide the established explanatory tooltip where required.

UI-004 Prototype inertia is not a lock

A control that is inert only because the prototype omits an interaction must not be documented or implemented as a business lock.

UI-005 Customer PDF and link parity

Customer Web Link and PDF use the same content model except where these requirements explicitly identify a presentation difference.

UI-006 Estimate table columns

Estimate and Change Order document tables use their normal item columns and do not show line-level percent complete.

UI-007 Tracked invoice table columns

Tracked invoice editors and applicable customer outputs show line-level billing progress columns according to their documented visibility rules.

UI-008 Preview actions after save

Invoice PDF preview and Customer web preview actions do not appear before invoice creation and are shown after save in their established location.

UI-009 Creation labels

Setup uses Continue only when another setup step follows and Create invoice when setup is complete. The editor uses Create invoice to persist the New Invoice.

UI-010 Linked to pattern

Invoice relationships use the established link identity pattern. Contract lineage belongs in Contract Details and not in Linked to.

UI-011 No internal ledger exposure

Customer documents may show resulting amounts and progress, but never expose Draft reservations, Discount Pool, Cost Plus ledger, Tax Credit ledger, Retainage ledger mechanics, or allocation rules.

UI-012 Revision specific acceptance

Estimate and pre-acceptance Change Order customer links identify one saved revision. An invalidated link explains that the document changed and directs the customer to the latest link.

UI-013 Current invoice emphasis

Customer Progress Invoice output emphasizes the current billed amount and visually de-emphasizes included lines with no current billing.

UI-014 PDF grayscale

The customer Progress Invoice PDF uses the established grayscale treatment while preserving the same financial content as the Web Link.

UI-015 No acceptance from incidental screenshots

Illustrative screenshots may clarify approved flows, but they do not create requirements that are absent from the written rules.

14 Internal UI Business Rules Requirements Index

ID	Requirement
UI-001	Conditional totals
UI-002	Internal Cost privacy
UI-003	Locked field clarity
UI-004	Prototype inertia is not a lock
UI-005	Customer PDF and link parity
UI-006	Estimate table columns
UI-007	Tracked invoice table columns
UI-008	Preview actions after save
UI-009	Creation labels
UI-010	Linked to pattern
UI-011	No internal ledger exposure
UI-012	Revision specific acceptance
UI-013	Current invoice emphasis
UI-014	PDF grayscale
UI-015	No acceptance from incidental screenshots

15 Canonical Numerical Examples**Example A Basic Progress Contract**

Before	Action	After
\$100,000 Accepted Estimate; no posted billing.	Open a \$40,000 Progress Invoice.	\$40,000 posted gross scope; \$60,000 Remaining subtotal; 40% Contract Progress; Partially Billed.

Example B Progress Contract with Discount

Before	Action	After
\$100,000 gross scope; \$10,000 Discount Pool;	Bill \$25,000 gross scope and apply \$2,500	\$75,000 Remaining subtotal; \$7,500 Discount

Before	Action	After
\$90,000 Contract Value.	discount.	Pool remains; \$22,500 Total Invoiced; Contract Value remains \$90,000.

Example C Cost Plus and Tax

Before	Action	After
\$10,000 taxable base; 10% taxable Cost Plus; 8% tax.	Bill 40% of the accepted scope.	\$4,000 base plus \$400 Cost plus fee equals \$4,400 Subtotal; tax is \$352; Total is \$4,752. Cost Plus is consumed once.

Example D Billed Contract Reopened

Before	Action	After
\$100,000 Current Contract and \$100,000 posted; status Billed.	Accept a positive \$10,000 Change Order.	\$110,000 Current Contract; \$100,000 history unchanged; \$10,000 Remaining subtotal; status Partially Billed.

Example E Payment Schedule Variance

Before	Action	After
\$100,000 Contract Value; 25% milestone plans \$25,000.	Create and open a valid \$20,000 Progress Invoice from that milestone.	Milestone actual is \$20,000; other percentage anchors do not rebalance; Contract availability decreases by the invoice's gross allocation.

Example F Retainage

Before	Action	After
\$25,000 gross Progress Invoice; 10% Retainage; no prior retained balance.	Open the invoice, then later open a \$1,000 partial Retainage release.	\$2,500 Total retained; \$1,000 Released; \$1,500 Held; original invoice records \$22,500 due; release does not increase Contract Progress.

16 Edge Cases

Feature	Case	Required result
Progress Invoice	Existing Draft	The Draft reserves availability; another tracked Draft is blocked.

Feature	Case	Required result
Progress Invoice	Partial source-line availability	The line remains eligible only up to its source allocation; aggregate Contract availability is also enforced.
Progress Invoice	Billed Contract reopened	A new positive Accepted CO creates new availability and changes the Contract to Partially Billed.
Progress Invoice	Locked posted invoice	An earlier or paid invoice remains valid for payments but cannot be financially edited.
Change Order	Negative CO after billing	Historical allocations remain; Contract may show Over invoiced and may generate Tax Credit.
Change Order	CO acceptance with Draft	Acceptance cancels the Draft, releases reservations, and requires a new invoice.
Change Order	Multiple Accepted COs	Each accepted revision accumulates into the same Contract and retains separate source identity.
Change Order	Discount exceeds prior pool	The CO may add the discount because it changes the contractual pool; the resulting pool cannot be negative.
Retainage	No Retainage	Tracking and release UI are absent.
Retainage	Fully invoiced with held balance	Contract Progress may be 100% while Retainage remains available for release.
Payment Schedule	Canceled milestone invoice	The live milestone returns to Planned; the invoice retains its historical milestone snapshot.
Payment Schedule	Accepted CO changes value	Future planned amounts recalculate; historical invoice actuals remain unchanged.
QBO	Transaction edited or paid externally	Cinderblock enters Sync conflict and prevents an unsafe automatic write.
QBO	Ambiguous create timeout	The integration reconciles by request identity and lookup before retrying.

17 Open Questions and Conflicts Found

The core Cinderblock Contract billing model is internally consistent. The following production accounting decisions remain unresolved in the authoritative QBO requirements and must be approved before implementation of the affected sync paths.

ID	Decision	Business behavior still required
OPEN-001	QBO line and account mapping	Define the QBO Product or Service and account mappings for Retainage, Retainage release, document discounts, and Cost Plus. The mapping must preserve each document total and line identity.
OPEN-002	QBO calculation order and rounding	Validate the exact QBO-compatible order and rounding for discounts, Cost Plus, Retainage, taxable negative lines, Tax Credit representation where applicable, and document totals under each supported QBO tax mode.
OPEN-003	Material QBO divergence	Define which QBO field changes create a material conflict and which metadata-only changes may merge without accounting risk.
OPEN-004	Keep QuickBooks version	Define how Keep QuickBooks version affects the Cinderblock accounting-document

ID	Decision	Business behavior still required
		representation and future sync baseline without importing QBO values into Contract billing ledgers.
OPEN-005	Safe overwrite void and detach	Define when overwrite, void, relink, detach, or leave-unchanged actions are permitted for transactions with payments, credits, deposits, closed periods, reconciliations, or tax constraints.
OPEN-006	Conflict permissions and support path	Define authorization, confirmation, audit, and support escalation for manual conflict resolution actions.
OPEN-007	Payment Schedule Remaining Balance shortcut	The source requirements refer to an existing Remaining Balance shortcut but do not define what value it inserts, when it is available, or how it behaves after Contract Value changes. Product must confirm the existing behavior before implementation.

Safe default until approval: preserve both systems, block the risky QBO write, surface the mismatch, and require intentional manual resolution.

18 Out of Scope

- Database schemas, migrations, persistence technology, and storage design.
- REST, GraphQL, API payload, service, class, or component architecture.
- Framework, state-management, queue, cache, webhook, polling, infrastructure, and deployment design.
- A new permissions model; existing Financial-area permissions remain authoritative unless a new action cannot map to them.
- Payment processor architecture and a second payment lifecycle.
- Unrelated CRM behavior or unrelated QBO integration redesign.
- Visual redesign beyond the states and conditional presentation required to communicate established business behavior.
- Automatic principal refunds, negative invoices, or credit memos for deductive Change Order over-invoicing in version one.

19 Technical Details

These constraints are included only because violating them would change observable business results.

- Use deterministic fixed-decimal currency arithmetic and one documented half-up rounding policy. Do not use binary floating point for persisted money.
- Allocate rounding remainders by stable source order so every displayed component reconciles exactly to its document total.
- Persist stable source-line identity, source document and revision, base or Cost Plus kind, accepted amount, taxability, supersession lineage, and allocation status.
- Persist an immutable tracked invoice billing-sequence key. Do not infer Previously Billed solely from mutable invoice dates or timestamps.
- Persist enough Contract revision information to reject stale Contract-affecting writes atomically and idempotently.
- Persist actual fractional quantities for accounting synchronization even when customer presentation hides them.
- Persist the last successful QBO concurrency value and material accounting snapshot needed to detect divergence before a destructive write.

Validation Requirements Index

ID	Requirement
VALID-001	Aggregate overbilling
VALID-002	Source line overbilling
VALID-003	Discount overuse
VALID-004	Existing Draft
VALID-005	Fully invoiced Contract
VALID-006	Change Order line conflict
VALID-007	Meaningless Change Order
VALID-008	Negative Contract Value
VALID-009	Schedule over-allocation
VALID-010	Blank milestone name
VALID-011	Invalid milestone allocation
VALID-012	Invoice exceeds milestone
VALID-013	Retainage unavailable
VALID-014	Retainage over-release
VALID-015	Final unused discount
VALID-016	Stale Contract write
VALID-017	Payment edit race
VALID-018	Stale acceptance link
VALID-019	QBO divergence
VALID-020	QBO retry ambiguity

20 Final Self Audit

Audit area	Result	Finding
Contract model	PASS	Progress is Contract-based from Estimate acceptance; Accepted COs revise the same Contract; source lineage remains intact.
Billing status	PASS	Contract billing status is distinct from CO lifecycle; Standard and Progress Invoice paths share Billed as the completed state.
Remaining subtotal	PASS	Defined as gross, pre-discount, source-backed availability after posted allocations and Draft reservations.
Feature interaction	PASS	Standard Invoice, Progress Invoice, CO, Retainage, Payment Schedule, Cost Plus, tax, and QBO boundaries are consistent.
Conditional customer display	PASS	Tax, Discount, Cost Plus, and Retainage are absent when inapplicable; internal Cost is never customer-facing.
Numerical examples	PASS	All six worked examples reconcile under the canonical formulas.
Requirement IDs	PASS	268 atomic requirements reviewed; 0 duplicate IDs found.
Screenshots	PASS	No rule depends solely on a screenshot. Written requirements remain authoritative.
Open decisions	PASS	Only unresolved QBO production accounting decisions are listed; no architecture questions are presented as business gaps.

Audit conclusion The specification is implementation-ready for the defined Cinderblock business model. QBO conflict resolution and accounting mappings remain intentionally blocked on the explicit decisions in Section 17.